C2 Bonding, Structure and Properties – Knowledge Organiser

AQA GCSE Chemistry • Set 1: Types of Bonding **Paper 1**

#	Question	Answer
1	Ionic bonding occurs between	Metal and non-metal
2	Covalent bonding occurs between	Non-metals
3	Metallic bonding occurs in	Metals
4	Ionic bond	Electrostatic attraction between oppositely charged ions
5	Covalent bond	Shared pair of electrons
6	Metallic bonding	Positive ions in a sea of delocalised electrons
7	What happens to electrons in ionic bonding	Transferred
8	Structure of an ionic compound	Giant ionic lattice
9	Formula of sodium chloride	NaCl
10	Formula of magnesium oxide	MgO

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AQA GCSE Chemistry • Set 2: States and Properties **Paper 1**

#	Question	Answer
1	Three states of matter	Solid, liquid, gas
2	State symbols	(s), (l), (g), (aq)
3	Meaning of (aq)	Dissolved in water
4	Limitations of the particle model	Particles shown as solid spheres; no forces shown
5	Why ionic compounds have high melting points	Strong electrostatic forces in all directions
6	When ionic compounds conduct	When molten or dissolved
7	Why they conduct then	Ions free to move
8	Why simple molecular substances have low boiling points	Weak intermolecular forces
9	Do simple molecular substances conduct?	No
10	Polymer	Very large molecule of many repeating units

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AQA GCSE Chemistry • Set 3: Giant Covalent Structures and Carbon **Paper 1**

#	Question	Answer
1	Examples of giant covalent substances	Diamond, graphite, silicon dioxide
2	Bonds per carbon in diamond	4
3	Why diamond is very hard	Every carbon has 4 strong covalent bonds
4	Bonds per carbon in graphite	3
5	Structure of graphite	Layers of hexagonal rings
6	Why graphite is soft	Layers slide (weak forces between layers)
7	Why graphite conducts	Delocalised electrons
8	Graphene	Single layer of graphite
9	Fullerenes	Hollow carbon molecules, e.g. C_{60}
10	Carbon nanotubes	Cylindrical fullerenes

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AQA GCSE Chemistry • Set 4: Metals and Alloys **Paper 1**

#	Question	Answer
1	Why metals conduct electricity	Delocalised electrons
2	Why metals conduct heat	Delocalised electrons
3	Why metals have high melting points	Strong metallic bonds
4	Why pure metals are soft	Layers of atoms slide easily
5	Alloy	Mixture of a metal with other elements
6	Why alloys are harder	Different sized atoms distort the layers
7	Example of an alloy	Steel
8	Why polymers are solid at room temperature	Large molecules, strong intermolecular forces
9	Bonds within polymer molecules	Strong covalent bonds
10	Why diamond does not conduct	No delocalised electrons

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AQA GCSE Chemistry • Set 5: Nanoparticles **Paper 1** **Triple only**

#	Question	Answer
1	Size of nanoparticles	1–100 nm
2	1 nm in metres	1×10^{-9} m
3	Size of fine particles ($PM_{2.5}$)	100–2500 nm
4	Size of coarse particles (PM_{10})	2.5–10 µm
5	Why nanoparticles behave differently	Very high surface area to volume ratio
6	Effect of side length ÷ 10 on surface area : volume	× 10
7	Uses of nanoparticles	Medicine, electronics, cosmetics, sun creams, catalysts
8	Use of silver nanoparticles	Antibacterial, e.g. wound dressings
9	Why less nanoparticle catalyst is needed	Larger surface area
10	Risk of nanoparticles	Unknown effects on health